

Week 12: Simple Linear Regression

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STA 220

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Overview

- We will be introducing simple linear regression
- Topics for this week
 - The simple linear regression model
 - Hypothesis testing for the slope of the regression line
 - Making predictions
 - Underlying assumptions for linear regression

height	weight
30	40
32	42
⋮	⋮

Working with 2 Variables

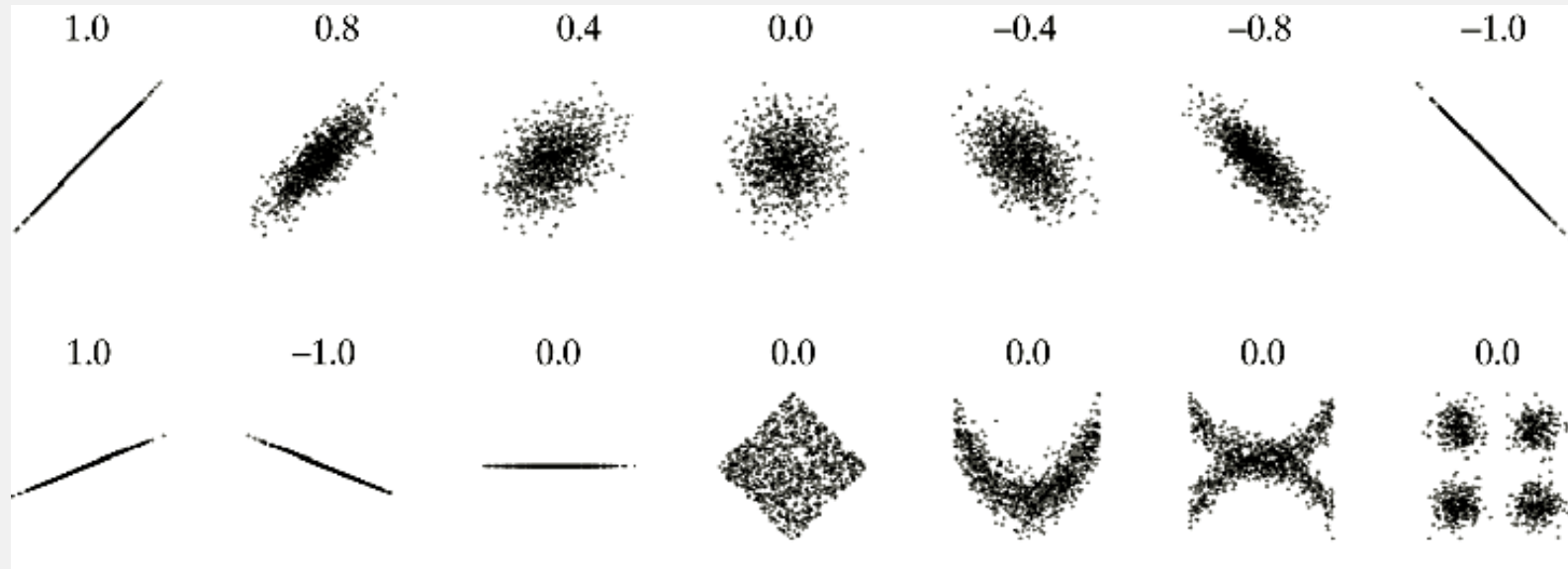
- We have previously seen in this course how to work with 2 variables
- In week 1, we saw how to visualize the relationship between 2 variables
 - 2 Quantitative Variables: Scatterplot
 - 2 Categorical Variables: Contingency table, Bar plot, etc.
 - 1 Categorical, 1 Quantitative: Side-by-side box plots, etc.
- Last week, we saw how to run a test to compare 2 groups
 - This is an example of examining the relationship between 1 quantitative variable and 1 categorical variable (with only 2 categories)

Working with 2 quantitative variables

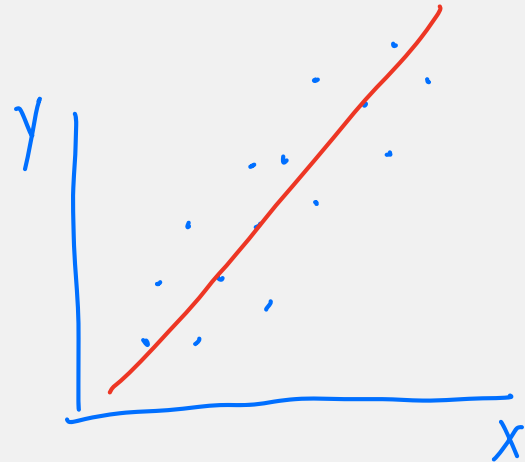
- We have seen that we can visualize the 2 quantitative variables using a scatterplot
- In addition, there are summary statistics to calculate:
 - **Covariance:**
$$Cov(X, Y) = \frac{\sum_i^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$
 - **Correlation:**
$$Cor(X, Y) = \frac{Cov(X, Y)}{SD(X)SD(Y)} = \frac{\sum_i^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i^n (x_i - \bar{x})^2 \sum_i^n (y_i - \bar{y})^2}}$$
- Correlation is a value between -1 and 1
 - 1 means perfect positive correlation
 - -1 means perfect negative correlation
 - 0 means no correlation

Correlation

- Note that correlation only measures linear relationships



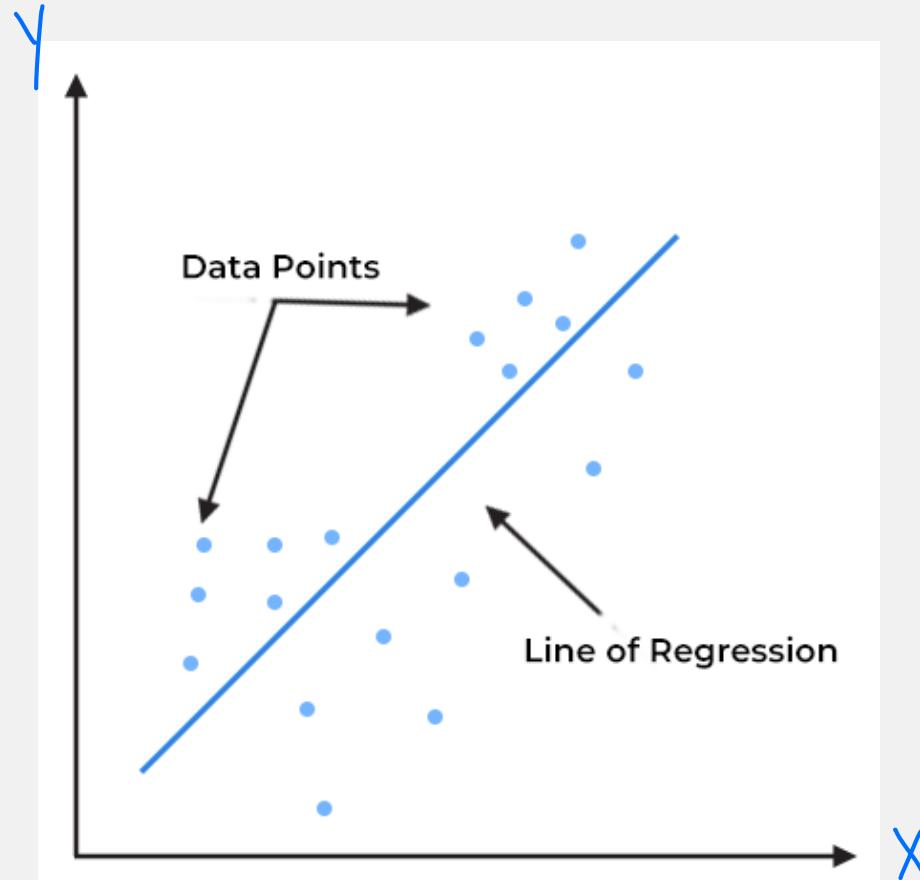
Simple Linear Regression



Two Quantitative Measurements

- Suppose we are looking at a scatterplot between two variables. If it makes sense to, we may want to summarize the relationship with a straight line.
- **Simple Linear Regression** provides a way to find the line that best summarizes the relationship between the two variables
 - we call Y the variable of interest or the response variable
 - this is because the values it takes will depend on the value of the other variable
 - our other X variable is called the explanatory variable or predictor
 - this variable is trying to explain the values we see in the response variable.

Summarizing with a Straight Line



The Regression Line

$$y = mx + b$$

- Since we want to summarize the relationship between X and Y with a straight line, we need to find the “best” line that fits through the data.
- Our **regression line** or **line of best fit** will take the form

$$y = b_0 + b_1x,$$

where

- b_0 is the **intercept** and represents what the value that y is expected to be when $x = 0$ (or where the line hits the y-axis)
- b_1 is the **slope** and represents the average change in y if x were to increase by 1 unit.

The Regression Line

- One important point is that the line cannot actually go through all the data points. But that's okay because the goal is to summarize the averages
- So the line only represents what will happen to Y on average for different values of X
- Given some data, how do we find the best line to put through the data? Well, we first need to define what we consider to be the “best” line.

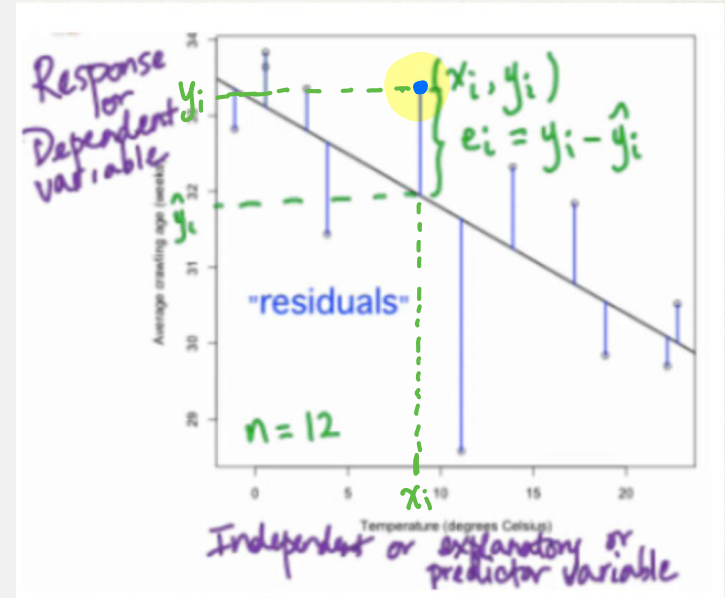
What defines the “best” line?

y vs. $\hat{y}$

- For a specific value of x , the line tells us that the **predicted value** for y is $\hat{y}$
- We can talk about how different the actual data value y is from the predicted one $\hat{y}$ by looking at the **residuals**

$$e_i = y_i - \hat{y}_i$$

- Among all the regression line candidates, we should choose the line where the residuals are the smallest.
 - The line should be one that sits in between all the data points and be at a minimum distance from all points



What defines the “best” line?

- For each regression line candidates, let's calculate a value that summarizes how large the residuals are:
 - Calculate residuals for all points
 - Since points can be both above and below the line, the residuals can be positive or negative, so let's square all the residuals
 - Then, sum up all the squared residuals.
 - In other words, we are calculating $\sum e_i^2$
- The line of best fit is the line that has the smallest $\sum e_i^2$

$$\min \sum e_i^2$$

Find the Slope and Intercept

- By minimizing the squared distance between all y values and their predicted $\hat{y}$ values, we find the following formulas for calculating the slope and intercept for the regression line
 - **Slope:** $b_1 = r \frac{s_y}{s_x}$ where s_y is the standard deviation of the response, s_x is the standard deviation of the explanatory variable, and r is the correlation between x and y
 - **Intercept:** $b_0 = \bar{y} - b_1 \bar{x}$, where we are working with the mean of y and x .
- So we only need 5 pieces of information to find the regression line equation.

Example: Credit Score

Suppose you want a better understanding of what goes into a good credit score. You manage to collect a random sample of annual income (in 100's of thousands) and Credit score values for 10 people.

Income (x)	1.3	1.1	0.8	1.2	1.4	0.9	0.9	1.4	1.2	1.0
Credit Score (y)	756	728	635	599	760	722	743	726	694	726

Summary Statistics: $Cov(x, y) = 3.16$, $s_x = 0.215$, $s_y = 52.573$,
 $\bar{x} = 1.12$, $\bar{y} = 708.9$

- a) What is the correlation coefficient?
- b) Compute the regression line
- c) Calculate the residual e_i for the first data point

$$a) \text{Cor}(X, Y) = \frac{\text{Cov}(X, Y)}{S_x S_y} = \frac{3.16}{0.215 \times 52.573} = 0.28$$

$$b) b_1 = r \frac{S_y}{S_x} = 0.28 \times \frac{52.573}{0.215} = 68.47$$

$$b_0 = \bar{y} - b_1 \bar{x} = 708.9 - 68.47 \times 1.12 = 632.21$$

Regression Line : $y = 632.21 + 68.47x$

$$c) (x_1 = 1.3, y_1 = 756)$$

$$\begin{aligned}\hat{y} &= 632.21 + 68.47(1.3) \\ &= 721.22\end{aligned}$$

$$e = y - \hat{y} = 756 - 721.22 = 34.78$$

How good is our line?

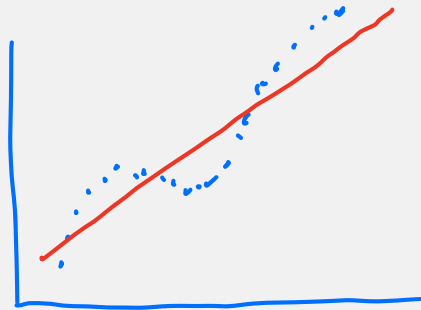
- The **coefficient of determination, R^2** , is a metric to determine how well the regression line fits the data

$$R^2 = \text{Cor}(X, Y)^2$$

- Its value is between 0 and 1.
- The R^2 represents the proportion of the variability in the response that can be explained with the regression line.
 - one way to look at this is that if the data don't seem to fit a linear relationship, then the line will have a hard time explaining the pattern in the data, so the R^2 will be small (think of a cloud of data).

How good is our line?

- We want R^2 to be large for the line to be good, but there is no cutoff for how large
- However you do need to be careful about using R^2 :
 - you can get large R^2 even when there is a non linear relationship
 - if the pattern in the data is not linear, then R^2 does not mean anything



Fitting linear regression in R

```
> income <- c(1.3, 1.1, 0.8, 1.2, 1.4, 0.9, 0.9, 1.4, 1.2, 1.0)
> credit_score <- c(756,728,635,599,760,722,743,726,694,726)
> model <- lm(credit_score ~ income)
> summary(model)
```

Call:
lm(formula = credit_score ~ income)

Residuals:

Min	1Q	Median	3Q	Max
-115.36	-15.78	22.88	31.01	49.13

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	632.38	94.50	6.692	0.000154 ***
income	68.32	83.01	0.823	0.434364

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 53.54 on 8 degrees of freedom

Multiple R-squared: 0.07805, Adjusted R-squared: -0.03719

F-statistic: 0.6773 on 1 and 8 DF, p-value: 0.4344

b_0

b_1

Hypothesis Testing for Linear Regression

Inference on the Slope

- We saw that the R^2 is used to check if the regression line captures the trend in the data well.
- But what if we wanted to see whether the linear relationship is statistically significant?
- We can do this by running a **hypothesis test on the slope** of the regression line.
 - this will give us an idea of whether we actually have the linear relationship our line claims, or if we are just seeing a relationship purely by chance.
- The hypothesis we are testing is about whether or not we have a linear relationship
 - specifically, that every time we increase x by one unit, y should change by b_1 units on average
 - if there is not linear relationship, then when x increases, y should on average not change at all
- so our hypothesis we are testing is H_0 : the true slope is 0.

b_1

Hypothesis Test on Slope

- Recall that hypotheses are always in terms of unknown population parameters.
- The population parameter for the slope is denoted as β_1 , which represents the **true** relationship or slope between X and Y
 - The slope that we calculate based on our sample, b_1 , is a sample statistic that we use to estimate β_1
- We write the test of the slope as $H_0: \beta_1 = 0$
- What would a scatterplot for X and Y look like if the true slope was 0?

Hypothesis Test on Slope

- We often use a two-sided test for the slope, so we test the alternative that a linear relationship exists.
- This gives the following:

$$H_0: \beta_1 = 0$$

$$H_A: \beta_1 \neq 0$$

- This can be described in words as:

H_0 : No linear relationship exists between x and y

H_A : There is a linear relationship

Hypothesis Test on Slope

- The test statistic is similar to what we have seen before:

$$\text{T-statistic} = \frac{b_1 - \beta_1}{SE(b_1)}$$

- You won't be asked to calculate the standard error
- We compare this test statistic to the T distribution with $df=n-2$
 - we find the p-value in exactly the same way as before, and make conclusions the same way too.

CI on Slope

- Just like with means, we can also make conclusions with confidence intervals.
- The confidence interval for the slope can be defined as:

$$\underbrace{b_1} \pm \underbrace{t_{n-2, \alpha/2}} \times \underbrace{SE(b_1)}$$

Using R

- Returning back to the previous example, the R output for linear regression gives us all the information we need to conduct the hypothesis test
- Use this information to write out the full hypothesis test on the slope

```
> income <- c(1.3, 1.1, 0.8, 1.2, 1.4, 0.9, 0.9, 1.4, 1.2, 1.0)
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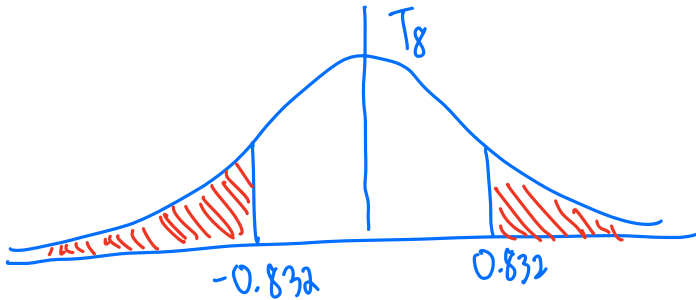
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$$H_0: \beta_1 = 0 \quad H_1: \beta_1 \neq 0$$

$$\text{Test Statistic} = \frac{b_1 - \beta_1}{SE(b_1)} = \frac{68.32 - 0}{83.01} = 0.832$$

Compare $T_{df=8}$



Using R , $p\text{-value} = 0.43$

$\Rightarrow$ Do not reject H_0 .

Using R

- It turns out that the test statistic and p-value for the two-sided test, and degrees of freedom are found directly in the R output

```
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$H_0: \beta_1 = 0$
 $H_A: \beta_1 \neq 0$

Making Predictions

Making Predictions

- Recall that $\hat{y}$ is the predicted value for a given x .
 - $(x_i, \hat{y}_i)$ is an ordered pair that actually sits on the line of best fit
 - (x_i, y_i) is an ordered pair in the data but won't necessarily sit on the line
- We can use the line of best fit to make predictions by calculating

$$\hat{y} = b_0 + b_1x$$

Credit score example

- The line of best fit is $\hat{y} = 632.38 + 68.32x$
- Find the predicted credit score if someone has an income of \$105,000.

$$X = 1.05$$

$$\hat{y} = 632.38 + 68.32 \times 1.05 = 704.1$$

- What about an income of -\$200,000?

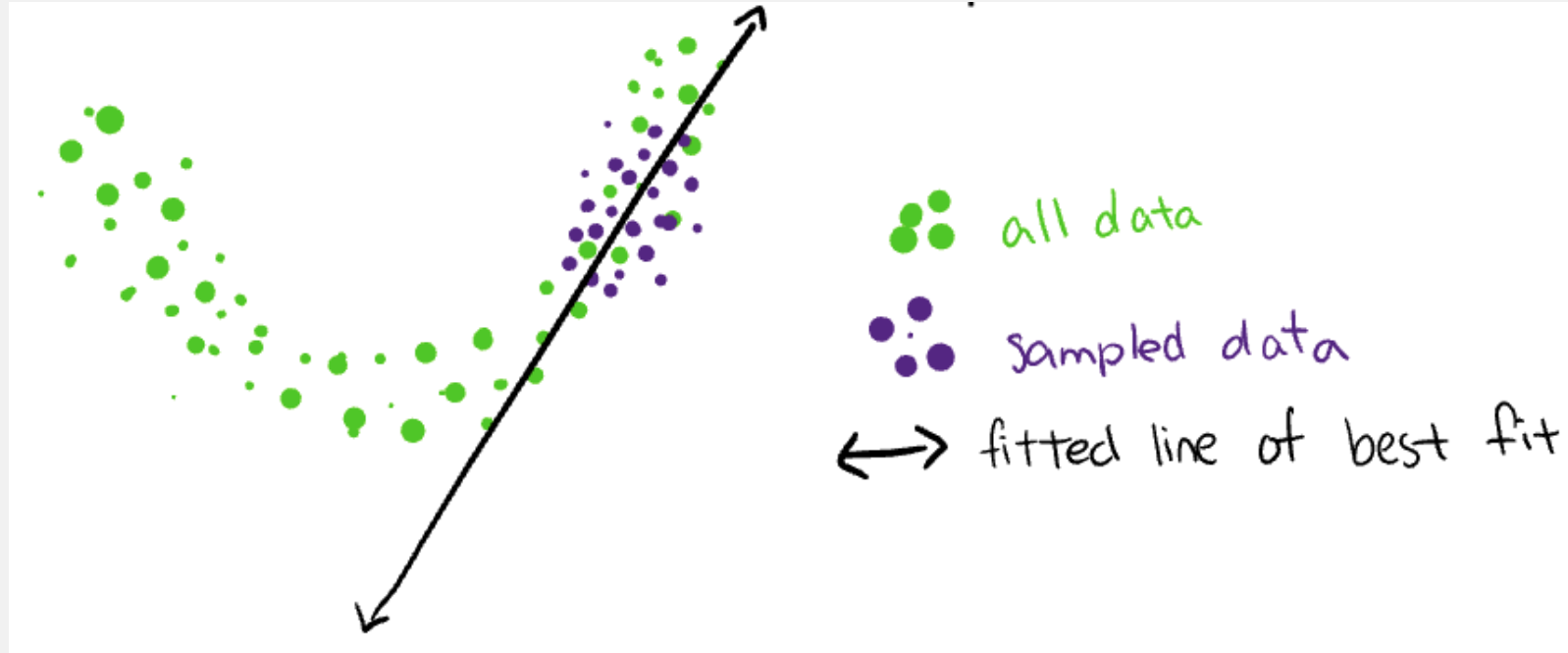
$$X = -2$$

$$\hat{y} = 632.38 + 68.32 \times (-2) = 495.7$$

Validity of Predictions

- It's important to think about whether the predictions we make are valid
- Credit scores range from 300 to 850, so any prediction outside of this range automatically don't make sense
- Negative incomes don't make sense either
- The range of data we used to fit the linear regression was \$80,000 to \$140,000. When making predictions, the X variable should stay within this range
 - If we go outside of this range, we are **extrapolating** the results

Risks of extrapolating



Assumptions

Assumptions of linear regression

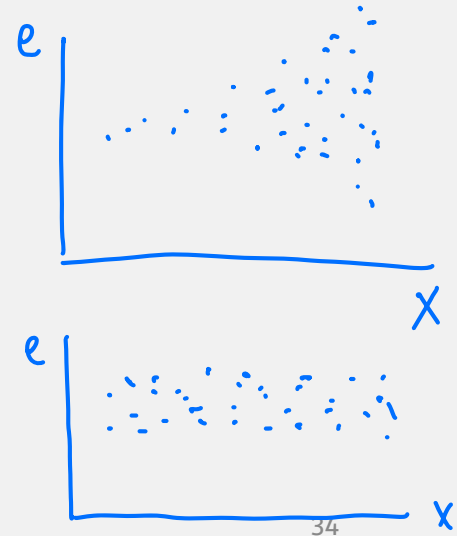
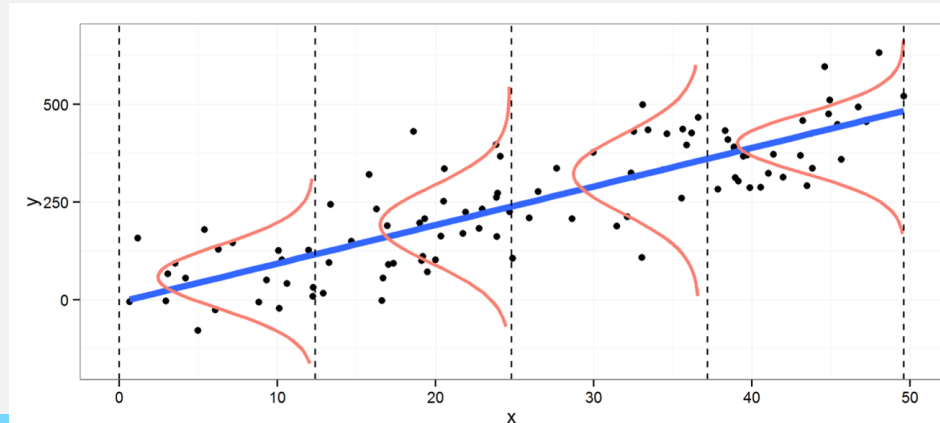
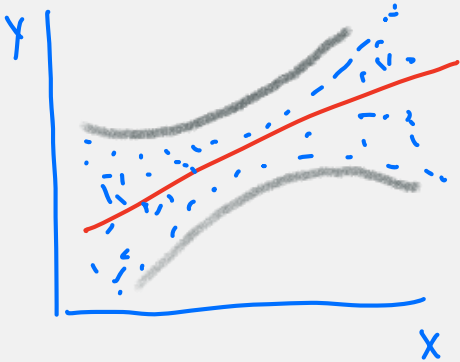
- As with all statistical procedures, we make certain assumptions which, when violated, means that we can't rely on our results.
- The assumptions in linear regression are:
 1. A linear model is appropriate (i.e. no curvature, no influential points, no groupings in the data)
 2. Observations are independent
 3. The variation in the error term is constant
 4. The errors are Normal.

Assumptions 1 and 2

- For assumption 1 (linear model is appropriate),
 - look at the scatterplot of the data
 - make sure there are no curve pattern and a straight line seems appropriate
- For assumption 2 (observations are independent),
 - no plot really helps here
 - in this case, you need to understand how the data were collected.
 - were they collected randomly?
 - if the data are not independent, then running a hypothesis test doesn't make sense because we need independence for it to work

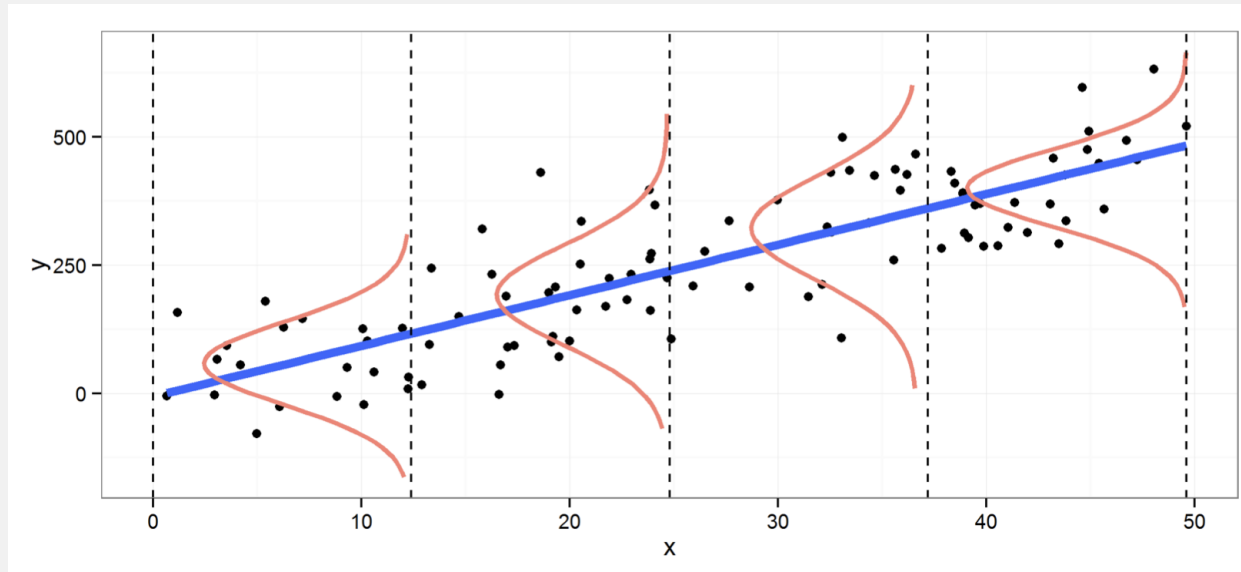
Assumptions 3 and 4

- The errors refer to the vertical distance between the data point and the regression line
- Assumption 3 (constant error variance) just means that each red curve must have the same variance
- need this assumption to be true in order for us to make inference about the slope



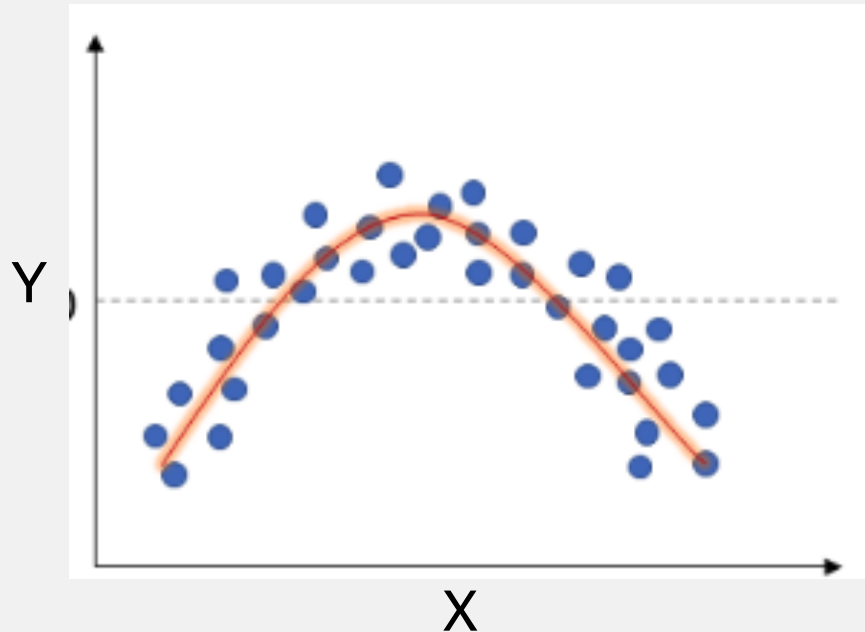
Assumptions 3 and 4

- Assumption 4 (errors are Normal): just says that the differences between the observed response and the line are Normal
 - A histogram of the errors should look bell-shaped



Example:

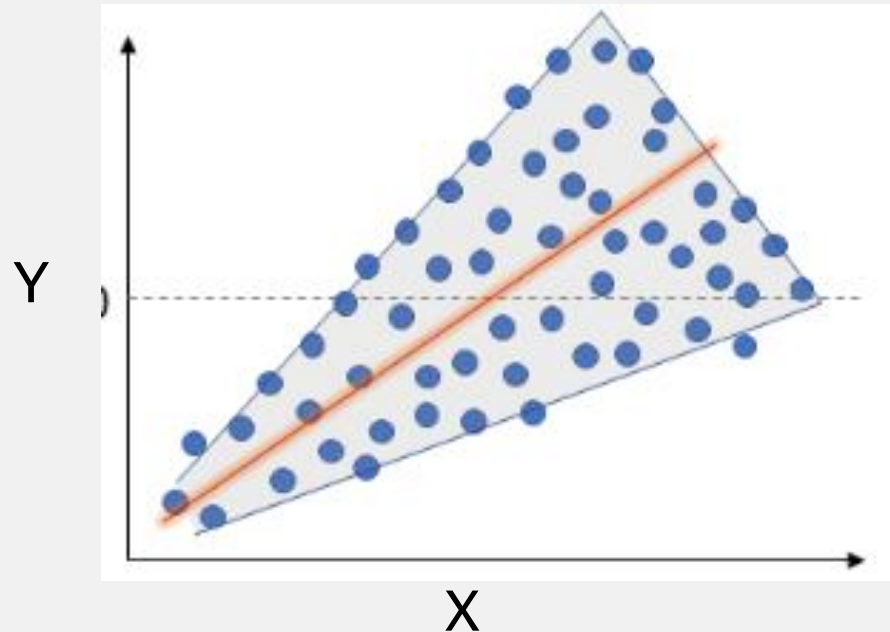
- Which assumption, if any, is violated based on this plot?



A1

Example:

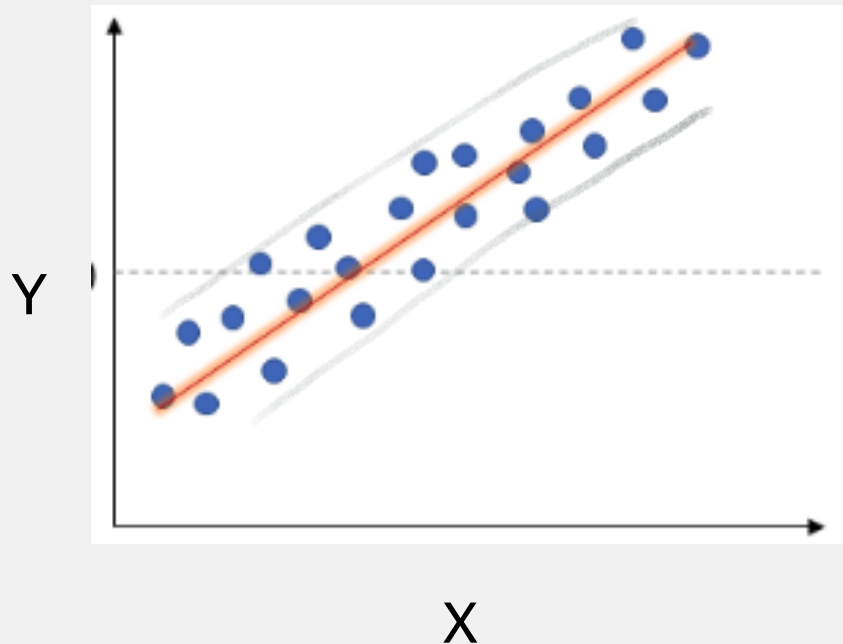
- Which assumption, if any, is violated based on this plot?



A3

Example:

- Which assumption, if any, is violated based on this plot?



A1 ✓

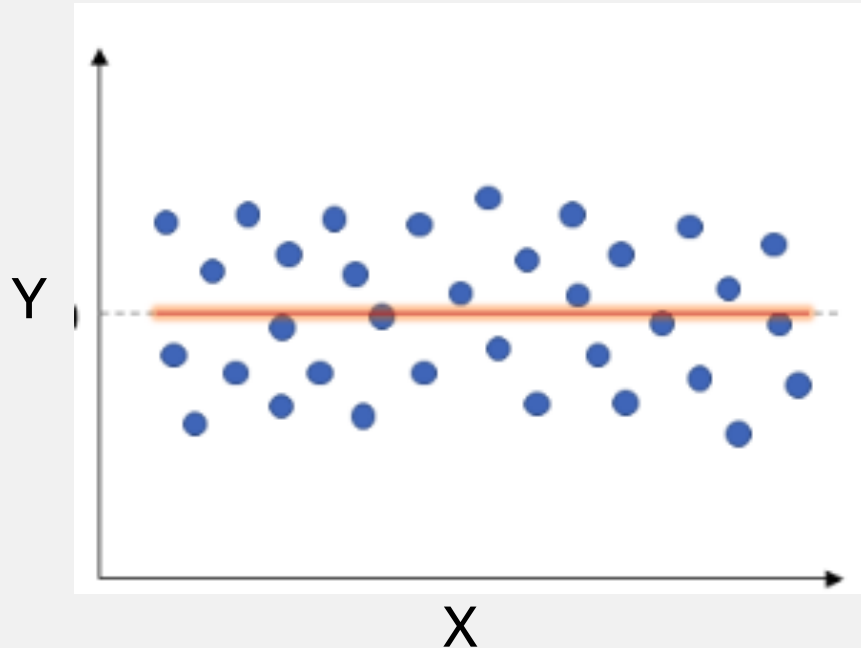
A2 ?

A3 ✓

A4: not the
right plot, but
no obvious problems

Example:

- Which assumption, if any, is violated based on this plot?



AI

Looking ahead

- We have made it to the end of the content for this course -- This is the last lecture of STA220
- Please see exam schedule for official final exam time and date
 - Announcement with official information will be posted soon!
- Pre-exam office hours will be posted
- Thank you for a great term!